
PrimeCore Minerals

LAN Port Security

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Introduction

The ability to protect a network from cyber threats is a range of measures that have been set up to allow users who need access the ability to access the data while also denying any unwanted access or damage. Sometimes this threat can be caused by an internal source and can be either intentional or deliberate.

Network points in the office or at site pose a cyber threat as they are direct links into the network and if port security isn't configured correctly, a whole site/office can be brought down by a user plugging in a dummy switch, which Duane has seen happen.

According to SentinelOne¹, 55% of insider breaches are attributable to employees who were careless or mistaken while accessing company resources. By securing the network ports, this could be reduced.

Port Security

By having a description for what is connected to that port recorded on the switch and labelling spare switches and placing them in a VLAN blackhole (VLAN that doesn't exist on that network, which for us is VLAN 1), network loops or potential access to company data is removed.

Our network has been configured so any spare ports will have a description of 'spare port' and will be tagged VLAN 1. As VLAN 1 doesn't exist on our network, the rogue PC can't connect anywhere.

The recommendation would be to administratively turn the ports off however it was left on in this demonstration to show the PC requesting a DHCP IP address but it not getting one.

Figure 1 shows the Rogue PC MAC address of 00:50:00:00:2a:00 on ens3 being turned on and it sending out a DHCP request, as per Figure 2.

```
QEMU (Rogue_PC)
Ubuntu 26.04 LTS srv-lnx-01 tty1
srv-lnx-01 login: labadmin
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Thu Oct  8 01:51:29 PM UTC 2026

System load: 0.09          Memory usage: 11%    Processes:      99
Usage of /: 40.2% of 13.67GB Swap usage:  0%    Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.

41 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

labadmin@srv-lnx-01:~$ sudo -i
[sudo: authenticate] Password:
root@srv-lnx-01:~# sudo netplan apply
[]
Cannot find unique matching interface for ens3
root@srv-lnx-01:~# sudo netplan apply
[]
Cannot find unique matching interface for ens3
root@srv-lnx-01:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:50:00:00:2a:00 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    altname enx005000002a00
    inet6 fe80::250:ff:fe00:2a00/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
root@srv-lnx-01:~#
```

Figure 1 – Rogue PC IP details including MAC address

```

23:52:17.298108 IP (tos 0xc0, ttl 64, id 0, offset 0, flags [none], proto UDP (17), length 313)
  0.0.0.0.bootpc > 255.255.255.255.bootps: BOOTP/DHCP, Request from 00:50:00:00:2a:00 (oui Unknown), length 285, xid 0x6f22276
  , secs 1, Flags [none]
    Client-Ethernet-Address 00:50:00:00:2a:00 (oui Unknown)
    Vendor-rfc1048 Extensions
      Magic Cookie 0x63825363
      DHCP-Message (53), length 1: Discover
      Client-ID (61), length 7: ether 00:50:00:00:2a:00
      Parameter-Request (55), length 12:
        Subnet-Mask (1), Default-Gateway (3), Domain-Name-Server (6), Hostname (12)
        Domain-Name (15), RP (17), Static-Route (33), NTP (42)
        URL (114), Unknown (120), Classless-Static-Route (121), Unknown (162)
      MSZ (57), length 2: 1472
      SLP-NA (80), length 0""
      Hostname (12), length 10: "srv-lnx-01"
23:52:18.026544 STP 802.1w, Rapid STP, Flags [Proposal, Learn, Forward], bridge-id 8000.4e:7c:a3:14:30:43.8002, length 36
  message-age 1.00s, max-age 6.00s, hello-time 2.00s, forwarding-delay 4.00s
  root-id 1000.26:13:23:6a:10:11, root-pathcost 2000, port-role Designated
23:52:18.504572 IP6 (flowlabel 0xf00ca, hlim 255, next-header ICMPv6 (58), payload length 16) fe80::250:ff:fe00:2a00 > ip6-allro
  uters: [icmp6 sum ok] ICMPv6, router solicitation, length 16
    source link-address option (1), length 8 (1): 00:50:00:00:2a:00
23:52:20.028240 STP 802.1w, Rapid STP, Flags [Proposal, Learn, Forward], bridge-id 8000.4e:7c:a3:14:30:43.8002, length 36
  message-age 1.00s, max-age 6.00s, hello-time 2.00s, forwarding-delay 4.00s
  root-id 1000.26:13:23:6a:10:11, root-pathcost 2000, port-role Designated
23:52:21.439229 IP (tos 0xc0, ttl 64, id 0, offset 0, flags [none], proto UDP (17), length 313)
  0.0.0.0.bootpc > 255.255.255.255.bootps: BOOTP/DHCP, Request from 00:50:00:00:2a:00 (oui Unknown), length 285, xid 0x6f22276
  , secs 4, Flags [none]
    Client-Ethernet-Address 00:50:00:00:2a:00 (oui Unknown)
    Vendor-rfc1048 Extensions
      Magic Cookie 0x63825363
      DHCP-Message (53), length 1: Discover
      Client-ID (61), length 7: ether 00:50:00:00:2a:00
      Parameter-Request (55), length 12:
        Subnet-Mask (1), Default-Gateway (3), Domain-Name-Server (6), Hostname (12)
        Domain-Name (15), RP (17), Static-Route (33), NTP (42)
        URL (114), Unknown (120), Classless-Static-Route (121), Unknown (162)
      MSZ (57), length 2: 1472
      SLP-NA (80), length 0""
      Hostname (12), length 10: "srv-lnx-01"
23:52:22.031256 STP 802.1w, Rapid STP, Flags [Proposal, Learn, Forward], bridge-id 8000.4e:7c:a3:14:30:43.8002, length 36
  message-age 1.00s, max-age 6.00s, hello-time 2.00s, forwarding-delay 4.00s
  root-id 1000.26:13:23:6a:10:11, root-pathcost 2000, port-role Designated
23:52:22.170958 IP6 (flowlabel 0xf00ca, hlim 255, next-header ICMPv6 (58), payload length 16) fe80::250:ff:fe00:2a00 > ip6-allro
  uters: [icmp6 sum ok] ICMPv6, router solicitation, length 16
    source link-address option (1), length 8 (1): 00:50:00:00:2a:00

```

Figure 2 – DHCP request from rogue PC. MAC address has been highlighted in green

Reference

1. SentinelOne 2026, Data Breach Statistics', SentinelOne

<https://www.sentinelone.com/cybersecurity-101/cybersecurity/data-breach-statistics/>